



The particles are also porous, so these soils are very moisture-retentive and rich in nutrients. However, clay soils can become waterlogged in wet conditions and form impenetrable crusts when dry. They are ideal for “hungry” plants like fruit trees.

Silt soils Pure silt soils are rare, usually occurring on river plains. They have a high nutrient content, but can become compacted and waterlogged like clay.

Loam With almost equal proportions of sand and clay, loam offers the best of all worlds, retaining enough water for plant roots to absorb, but allowing excess moisture to drain away and preventing waterlogging. It also holds on to nutrients well, making it the perfect garden soil for most plants.

Improving your soil

Whether you have a dry, sandy soil or a sticky clay one, your plants will grow better if you improve its quality by applying plenty of organic matter, such as well-rotted farmyard manure, garden compost, or spent mushroom compost. Organic matter coats sandy soil particles, helping them to retain more water, while it also opens up the structure of clay soils, allowing water to drain more easily.

Either dig it into the soil when preparing for planting or lay a thick layer as a mulch.

ABOVE Plants that thrive in dry, sandy soil

Dry, sandy soils and a sunny site provide the perfect conditions for many euphorbias, Californian poppies, eremurus, and succulents, such as agaves.

Worms and microorganisms will then work it into the soil.

The structure and drainage of heavy clay soils can also be improved by the addition of horticultural grit. Simply dig the grit into the soil over a large area. In extreme circumstances, very heavy, waterlogged soils may require drains.

Understanding pH

The pH of a soil is a measure of its acidity or alkalinity, which also influences the types of plants you can grow successfully. It is measured on a scale from 1 to 14; neutral soil has a pH of 7, a number below this indicates an acid soil, while alkaline soils have a pH above 7. A precise measurement can be obtained by performing a simple soil test (see right). Although many plants are tolerant of a wide pH range, there are some that are adapted to a particular soil type. Rhododendrons, azaleas, and heathers (*Erica*), for example, require an acidic soil, while lilacs (*Syringa vulgaris*) and the pineapple broom (*Cytisus battandieri*) grow best in alkaline soil.

TESTING YOUR SOIL

An easy way to find out what type of soil you have is to dig up a small sample and roll it between your fingers to feel the texture. Soils with a high water content can be rolled into a ball. You can also use a simple kit to test the pH.



Sandy and silty soil

Soils rich in sand feel gritty between your fingers; silty soils feel silky. Both fall apart when rolled into a ball. Sandy soils are easy to dig, and warm up quickly in spring. You may also find that borderline hardy plants survive winters better in these dry soils.



Clay soil

Smooth, sticky and dense, clay soils retain their shape when rolled into a ball, and those with a very high clay content remain intact even when formed into a horseshoe. They are often described as “heavy” soils because they are difficult to dig.



Using a pH testing kit

You can buy pH testing kits online and from garden centers. Place a small soil sample in the tube and add the solution provided with the kit. Wait until the solution changes color, and then match it to the chart. Take a few readings from different areas of the garden, as they may have different pH values.